



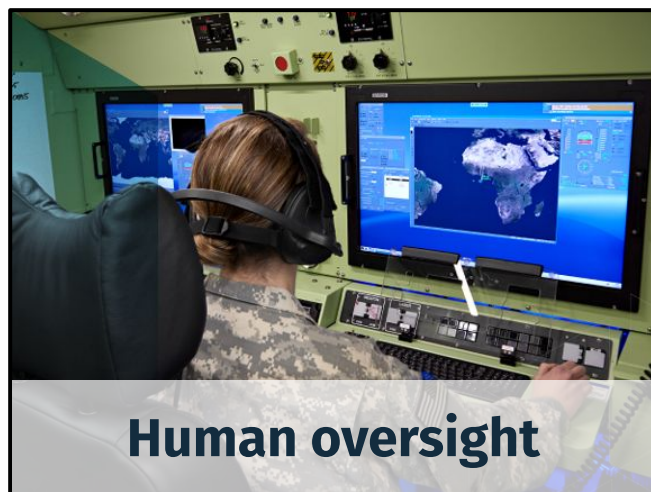
Spy Turtle: Double Vision

Ali Baharanyi: AI Technician
Benjamin Conger: AI Technician
Crissy Cabrera: AI Technician

Jiwoong Chang: AI Technician
Ruben Blot: AI Technician
Steve Gutierrez: AI Technician

Carnegie
Mellon
University

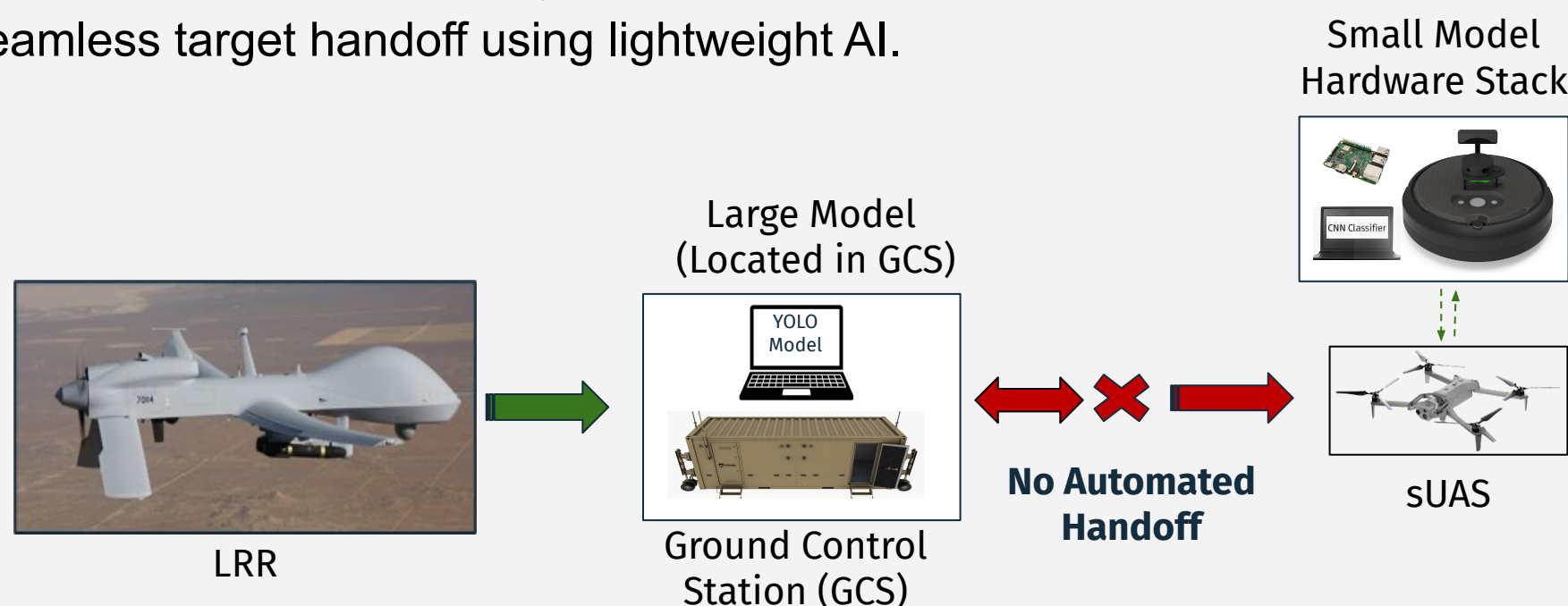
Enhancing Unmanned Aerial System (UAS) capabilities through AI-powered object recognition and automatic target handoff between large and small drones.



* Real-world military scenarios. * Autonomous AI target recognition. * Improved operational efficiency.

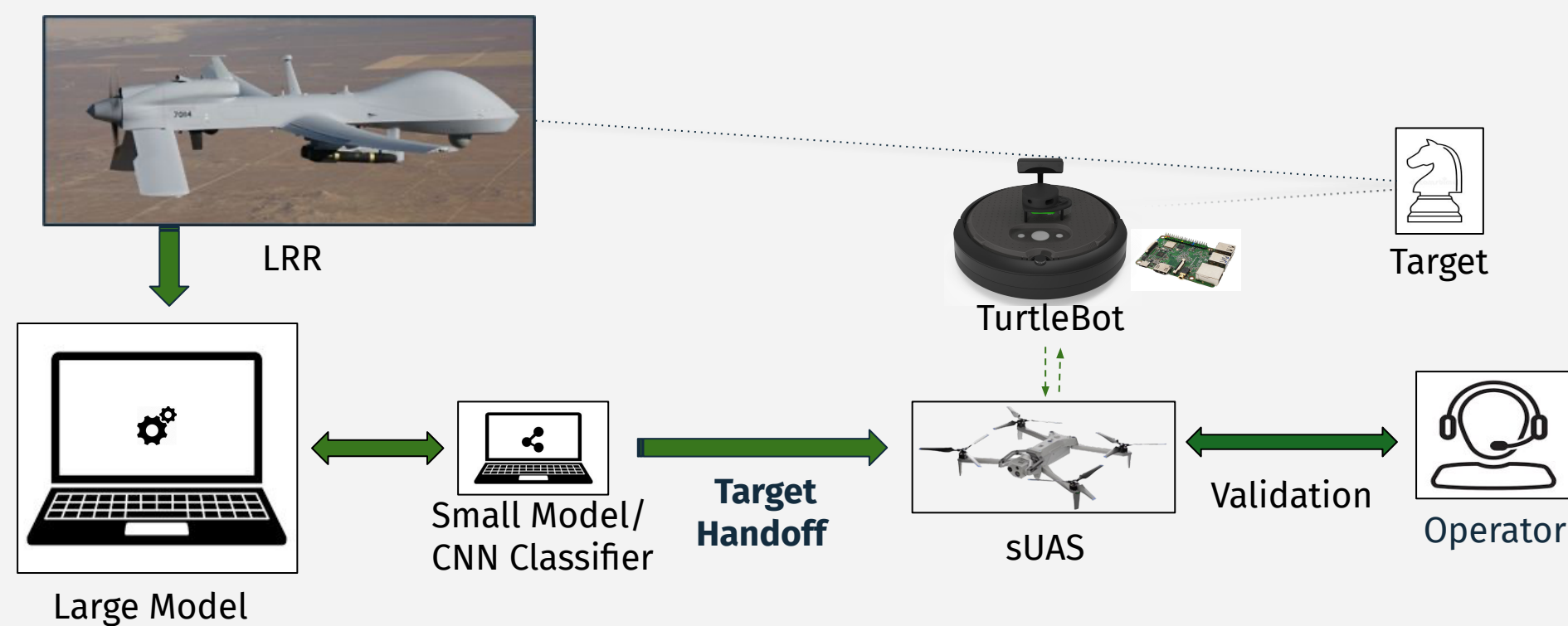
Problem Definition

- Capability Gap:** Strategic drones use advanced AI; tactical drones rely on manual methods.
- Operational Risks:** Manual handoff increases delays and operational risks.
- Computational Constraints:** Small drones limited by hardware, restricting onboard AI capabilities.
- Workflow Inefficiency:** Manual processes hinder fast decision-making.
- Proposed Solution:** Deploy "Double Vision" AI to automate seamless target handoff using lightweight AI.



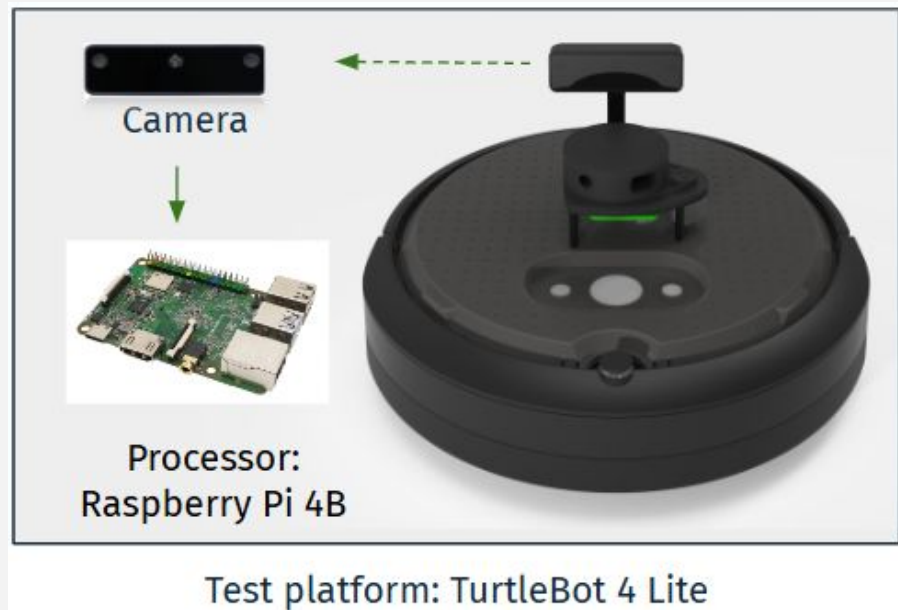
System Workflow

- Large drone captures wide-area imagery.
- YOLO model identifies and crops potential targets.
- CNN model validates targets onboard the small drone.
- Ground operator confirms results.
- Creates a continuous, streamlined data flow.
- Integrates human oversight at crucial verification stages.



Project Goals

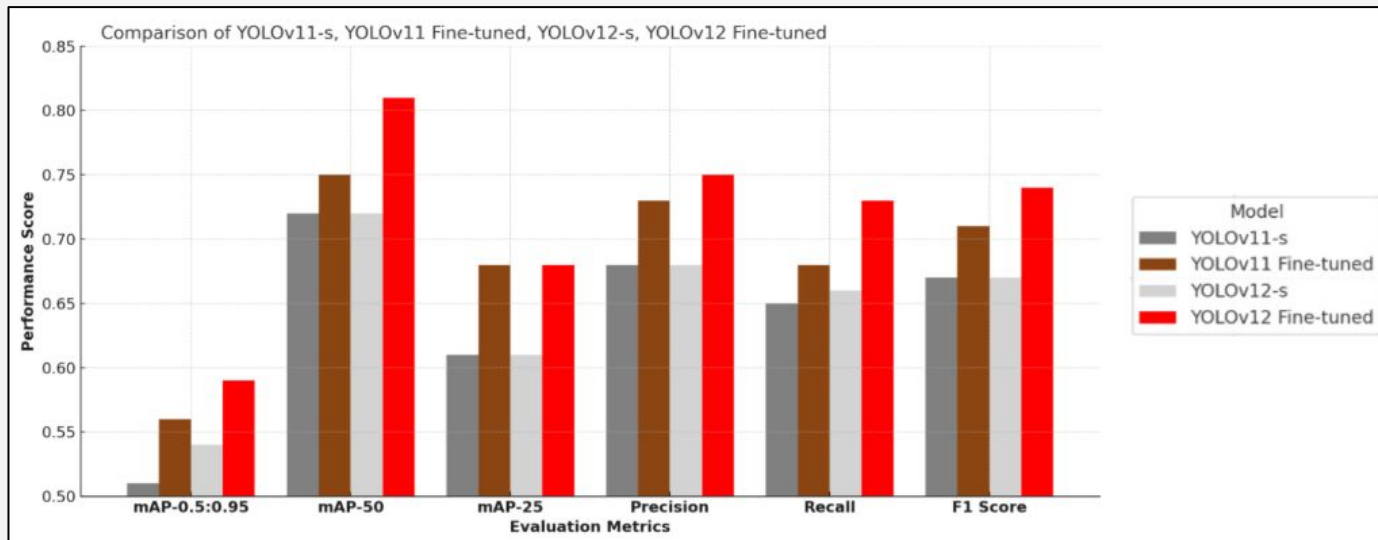
- Deploy real-time AI on tactical drones.
- Automate rapid drone-to-drone target handoff.
- Optimize for low-power tactical hardware.
- Enhance speed, reduce human errors.
- Validate performance in realistic conditions.
- Provide feasibility on hardware-limited platforms (Raspberry Pi).
- Reduce decision-making latency for tactical operations.



Technical Approach

Large Model (YOLO)

- Custom YOLO model trained on extensive drone images.
- Labeled positive (target) and negative images for clarity.
- Roboflow labeling combined with K-means clustering improves accuracy.
- Focus on selecting high-quality, representative images.
- Reduced computational demands by optimizing dataset size.
- Evaluated and fine-tuned model significantly improves detection metrics (mAP, precision, recall, F1-score)



YOLO Versions and Fine-Tuning

Small Model (CNN)

- MobileNetV2 trained specifically on YOLO-cropped images.
- Lightweight model optimized for rapid, tactical inference.
- High accuracy with ROC AUC = **0.91**, outperforming baseline methods.
- Successfully validated in realistic tactical mission scenarios.
- Directly deployable onto tactical drone hardware.
- Consistent performance in diverse operational environments.

Achievements

Model Integration:

- Streamlined large & small AI models integration.

Performance Metrics:

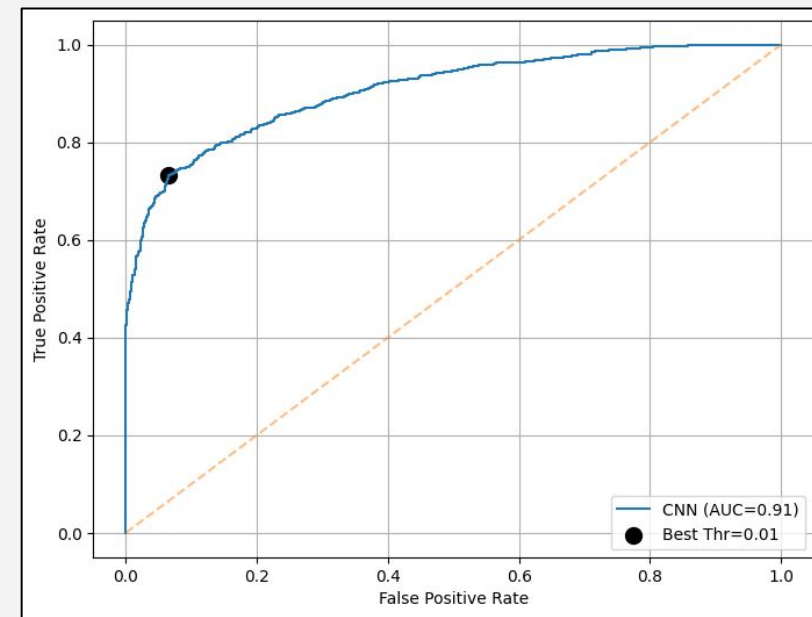
- ROC Curve (CNN AUC=**0.91**).
- Precision-Recall Curve (AUC=**0.94**).
- Optimal CNN Threshold = 0.014.

Operational Validation:

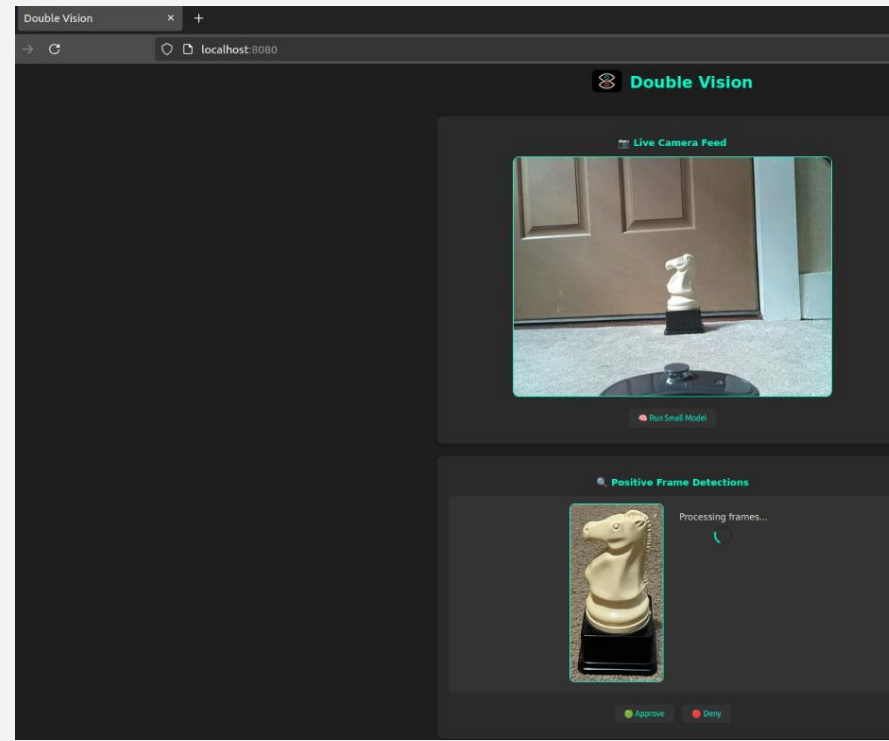
- Reliability confirmed on Raspberry Pi hardware.
- Outperformed manual target identification.

User Interface:

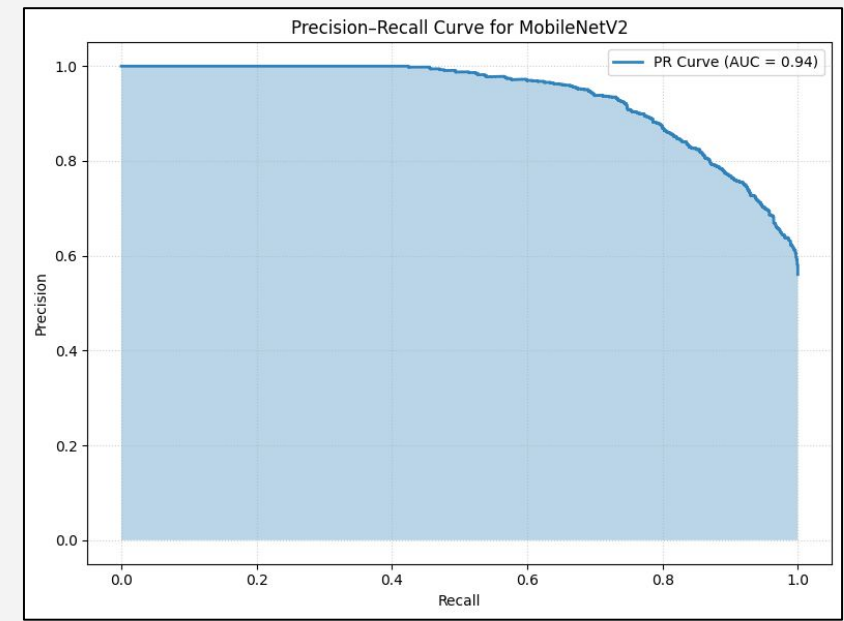
- Flask App for live monitoring and drone operation.



ROC Curve (CNN AUC=**0.91**)



Double-Vision Flask App Interface



Precision-Recall Curve (AUC=**0.94**)

Future works and Applications

- Real-world drone deployment.
- Integrate advanced sensors (LiDAR, thermal).
- Expand to complex environments (urban, forest).
- Scale solution for drone swarm coordination.
- Leverage multi-sensor fusion for robustness.
- Explore advanced AI scaling for multi-drone tasks.
- Extend to broader operational and geographic scenarios.
- Test operational limits in highly dynamic missions.

